

FINANCIAL OPERATIONS INTELLIGENCE AGENT

FitzSight

面向金融运营的调查 Agent。它从 KPI 异常出发，调用确定性工具组织证据，再把有边界的结论交给判断人。

KPI CHANGE INVESTIGATION EVIDENCE ID HUMAN DECISION

THE UNFINISHED JOB

FitzSight 的应用场景

当团队看到 KPI 异常，却还要在 Dashboard、SQL、事件日志和文档之间来回核对时，FitzSight 会把这些步骤串成一条可追溯的调查链。

DASHBOARD	SQL	PYTHON	EVENT LOG	DOCUMENT	REPORT
-----------	-----	--------	-----------	----------	--------



TASK DIAGNOSIS

为什么需要 Agent

调查会根据中间结果改变下一步。Agent 只在行动目录和证据边界内编排分支，数字仍由确定性工具计算。

01 / DASHBOARD

预定义观察

回答“发生了什么”，但不会为一次新调查编排统计、贡献、事件和文档核验。

固定视图

02 / CHAT WITH CSV

生成式解释

容易把数字计算和解释权同时交给模型；证据、权限与停止条件不稳定。

模型拥有解释权

03 / FITZSIGHT

受限调查

意图先门控；结果驱动批准分支；确定性工具计算；证据不足则停止归因。

系统拥有边界

04 / AGENT TEST

下一步由状态决定

行动目录、Evidence 状态与 Verifier 共同决定继续、降级或拒绝输出。

可检查的闭环

BOUNDARIES BEFORE AUTONOMY

FitzSight 如何完成调查

系统先判断问题是否在批准范围内，再生成受限计划、调用确定性工具并核验证据。证据不足时，流程停在人工确认。

- | | |
|----|---|
| 01 | Local intent gate
目录外问题在模型调用前拒绝 |
| 02 | Constrained plan
仅批准 intent / action 顺序 |
| 03 | Deterministic tools
SQL / statistics / decomposition |
| 04 | EvidenceClaimVerifier
核验证据、状态与因果措辞 |



UNSUPPORTED INTENT → TOOL ERROR → MISSING CORROBORATION →

INSUFFICIENT_EVIDENCE / HUMAN REVIEW

FitzSight v0.12 — Verified Hero Run Evidence

Brokerage / FinTech Operations Analyst
Autonomous investigation. Human decision.

USER QUESTION
Why did European FTD conversion deteriorate after July 15?
VERIFIER: PASS | 6/6 claims verified

BOUNDED ADAPTIVE EXECUTION TRACE

- P1 inspect_schema [executed]**
Required operational fields were inspected.
Evidence: E0002
- P2 query_affected_cohort [executed]**
Affected Europe Team A+B cohort measured.
Evidence: E0003
- P3 query_control_cohort [executed]**
European control cohort measured.
Evidence: F0004
- P4 statistical_validation [executed]**
Conversion and response-time shifts tested.
Evidence: E0005, E0006
- P5 contribution_decomposition [executed]**
Team contribution decomposition completed; result controls the next approved branch.
Evidence: E0009, E0010
- P6 anomaly_scan [executed]**
Approved by contribution/statistical branch.
Evidence: F0011, F0012
- P7 event_check [executed]**
Operational event context checked after the latency/statistical signal.
Evidence: E0013
- P8 document_evidence_check [executed]**
Synthetic routing ticket paragraph linked by stable source reference.
Evidence: E0014
- P9 evidence_boundary [executed]**

QUESTION

Why did European FTD conversion deteriorate after July 15?

planner_mode = deterministic_fallback · 截图来自真实本地 runtime

APPROVED STEPS

9

schema → compare → stats → contribution
→ anomaly → event → document

EVIDENCE REGISTRY

18

最终 Registry 记录

FOLLOW-UP

VERIFIED

verified_with_guardrail

PRODUCT DEMO / DETERMINISTIC HERO

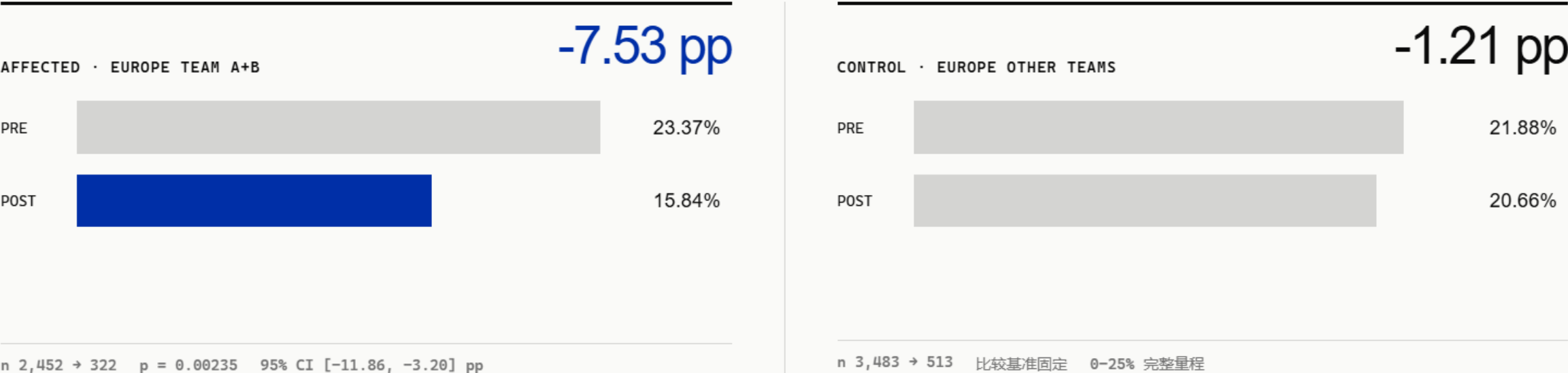
欧洲 FTD 异常调查

示例从一个业务问题出发，依次完成九个获准步骤，并把每一步的输入、结果和 Evidence ID 留在运行轨迹中。

AFFECTED VS CONTROL

欧洲 FTD 异常有多大

7 月 15 日后，受影响团队的 FTD 下滑 7.53 个百分点，对照组同期仅下滑 1.21 个百分点。延迟指标也同步恶化。



RESPONSE MEDIAN

94.30 → 123.45 min

+29.15 min

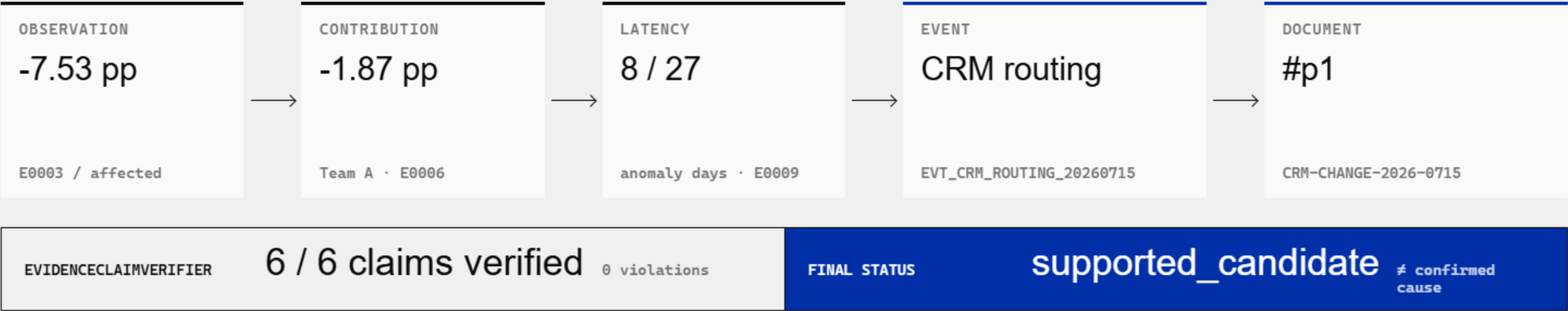
Mann-Whitney p = 1.86e-17

SYNTHETIC BENCHMARK · 百分点 (pp) 与百分比 (%) 已区分 · 显著性不等于因果性

SUPPORTED CANDIDATE, NOT PROVEN CAUSALITY

证据指向什么

多源证据共同指向 CRM routing，但当前强度只支持“候选原因”。系统保留这一边界，不把相关性写成已证实因果。



TRUST REQUIRES A STOP CONDITION

什么时候应该停止归因

事件时间接近并不足以建立因果关系。缺少测量驱动或文档佐证时，FitzSight 会拒绝归因，并把结果交给人工复核。

A FALSE-CORRELATION REFUSAL

诱人故事被证据边界截断

OBSERVED	Asia FTD -8.13 pp
NEARBY EVENT	Office relocation
MEASURED DRIVER	Affiliate -15.81 pp

CAUSAL EVIDENCE

NO · REFUSED

p = 0.00463 · 4/4 claims verified

B FAIL-CLOSED PATH

依赖失败不等于允许猜测

DEPENDENCY	tool_error_fail_closed
DOCUMENT	missing
CAUSAL SUPPORT	insufficient
ATTRIBUTION	withheld
VERIFIER	PASS

FINAL ANSWER

BOUNDED · VERIFIED

一个 unseen CRM seed 同样返回 insufficient evidence

REUSABLE INVESTIGATION GRAMMAR

哪些问题可以复用这套方法

CRM、净入金、客户行为、营销线索和伪相关调查共用一套调查语法，差别只在获准动作和确定性分析工具。

QUESTION	PLAN	TOOLS	EVIDENCE	VERIFY
APPROVED INTENT	DETERMINISTIC ANALYSIS PATTERN		SYNTHETIC EVIDENCE	
CRM / FTD	affected-control · stats · event/document			Hero verified
Net deposits	deposit/withdrawal decomposition · concentration			5 claims · top-11 91.6%
Customer intelligence	behavior features · descriptive segmentation			6,770 · 100% coverage
Marketing lead quality	volume · mix · within-channel performance			+315.0% · -10.84 pp
False correlation	nearby event · falsification · causal guardrail			4/4 · rejected

5 / 5

fixed synthetic scenarios PASS

复用的是 boundary + action catalog + tools + Evidence Registry + Verifier

PLANNER OUTPUT = UNTRUSTED INPUT

DeepSeek 在系统中负责什么

DeepSeek 只负责选择获准的 intent 和 action 顺序。权限控制、SQL、数值计算和结论核验都留在本地。



网络调用前拒绝目录外问题

Flash / Pro 白名单 · JSON purpose

只读 SQL · 统计 · 分解 · falsification

Evidence · digest · status · causal wording

EVIDENCE REGISTRY		每个数字、状态与主张都有稳定 Evidence ID	
PLANNER CAN	选择批准 intent / action 顺序	PLANNER CANNOT	生成任意 SQL · 计算数字 · 执行金融动作 · 决定最终结论
PROVIDER EVIDENCE	Mock contract / timeout / JSON / truncation / telemetry redaction PASS	LIVE STATUS	DeepSeek live = not_requested

ARCHITECTURE ABLATION, NOT GENERIC LLM BENCHMARK

系统经过了哪些验证

固定场景、对抗测试和留出测试检查路由、证据与拒绝行为；消融结果用于观察 Verifier gate 对不安全输出的影响。

FIXED BENCHMARK



5 / 5

evidence coverage 100%

ADVERSARIAL



8 / 8

all categories PASS

HOLDOUT ROUTE / VERIFY / EVIDENCE / REFUSAL



8 / 8

2 unseen seeds × 2 workflows × 2 paraphrases

SUPPORTED_CANDIDATE RATE



75%

1 unseen CRM seed → insufficient evidence

FULL FITZSIGHT

0%

unsafe adversarial answers

Verifier gate ON

NO-VERIFIER GATE

100%

unsafe adversarial answers

same architecture fixtures

SYNTHETIC PROTOCOL ONLY · 75% 不是“真实根因准确率” · 消融比较的是架构门控，不是 FitzSight vs Generic LLM

EVIDENCE-GROUNDED FINANCIAL OPERATIONS INVESTIGATION

当前进展 与下一步

PoC 已开源并可按文档复现。
真实部署仍需企业级控制。

当前版本完成了有界调查、确定性工具和证据核验。下一步补齐 SSO、RBAC、隐私隔离与生产监控。

IMPLEMENTED / VERIFIED	REUSABLE	PRODUCTION BLUEPRINT
5 个批准 intent 确定性工具与 Verifier 98 tests PASS Streamlit HTTP 200 / ok	MIT source code synthetic generator benchmark catalog docs + deterministic fallback	Enterprise SSO / RBAC PII masking tenant isolation / retention production observability

DELIBERATE PRODUCT BOUNDARY

可调查、测量、比较、检验、追溯；不交易、不转账、不授信、不替代授权金融决策者。

证据先行，结论有边界，决策由人负责。